1. Why do we need an activation function in a neural network? What happens if we don't use one?

2. Explain why sigmoid is called a "squashing function".

3. Why is ReLU more popular than Sigmoid in deep neural networks?

4. What problem does ReLU solve compared to Sigmoid?

5. When should we use Softmax instead of Sigmoid?

6. Why is Binary Cross Entropy preferred over MSE for classification?
7. If true label y = 1 and predicted probability $\hat{y} = 0.9$, calculate the Binary Cross Entropy loss.
8. If y = 0 and $\hat{y} = 0.8$, calculate the loss.
9. Compute sigmoid(0), sigmoid(2), and sigmoid(-2).